

Размеры слоев

вход

выход

каналы

активация

Conv 1 7×7 S2

5/2

32

85^2

~~Conv 2~~

Max pool 3×3 S2

5/4

32

25^2

Conv 2 5×5

5/4

~~64~~ 64

45^2

Conv 3 3×3 S2

5/8

128

25^2

Conv 4 1×1

5/8

~~256~~ 256

45^2

Conv 5 3×3 S2

5/16

256

5^2

Conv 6

1×1

5/16

512

25^2

$$FLOPS = 2B \cdot S^2 \cdot C_{out} \cdot C_{in} \cdot K$$

2B - jumlah core

S^2 - ukuran kernel

C_{out} - jumlah barisan

C_{in} - jumlah kolom

K - jumlah operasi

Core	Baris	FLOPs
conv1	$2(5/2)^2 \cdot 32 \cdot 3 \cdot 96$	2352
conv2	$2(5/4)^2 \cdot 64 \cdot 32 \cdot 25$	5400
conv3	$2(5/8)^2 \cdot 128 \cdot 64 \cdot 9$	2304
conv4	$2(5/16)^2 \cdot 256 \cdot 128$	1024
conv5	$2(5/16)^2 \cdot 256^2 \cdot 9$	4608
conv6	$2(5/16)^2 \cdot 512 \cdot 256$	1024
fc1	$2 \cdot 512 \cdot 256$	262144
fc2	$2 \cdot 256 \cdot 100$	51200

$$FLOPS (S, B) = 8 (17372 S^2 + 373844)$$

Beca

$$conv: 4704 + 51200 + 73728 + 32768 +$$

$$+ 585824 + 137072 = 885296$$

$$fc1: 512 \cdot 256 \cdot 256 = 131328$$

$$fc2: 256 \cdot 100 \cdot 100 = 25700$$

$$1040324$$

Bytes Moved - (Kalkulasi jumlah GPU memory in ram)

1) Garis-garis input & output memory

$$input: 3 \cdot S^2$$

$$broad conv1: 32 \cdot (5/2)^2 = 32 \cdot 8 = 256$$

$$broad pool: 32 \cdot (5/4)^2 = 256$$

$$broad conv2: 64 \cdot 5^2 = 1600$$

$$broad conv3: 128 \cdot 5^2 = 3200$$

$$broad conv4: 256 \cdot 5^2 = 6400$$

$$broad conv5: 256 \cdot 5^2 = 6400$$

$$broad conv6: 512 \cdot 5^2 = 12800$$

$$broad pool: 512$$

Basead FC₁ = 256

Basead FC₂ = 100

21 ЧТЕНУЕ и ЗАПИСЬ

Алго	Учел	Занес
Conv 1	35 ²	85 ²
relu	85 ²	85 ²
pool	85 ²	25 ²
Conv 2	25 ²	45 ²
relu	45 ²	45 ²
Conv 3	45 ²	25 ²
relu	25 ²	25 ²
Conv 4	25 ²	45 ²
relu	45 ²	45 ²
Conv 5	45 ²	5 ²
relu	5 ²	5 ²
Conv 6	5 ²	25 ²
relu 6	25 ²	25 ²
pool	25² 25 ²	5 ²
FC 1	25² 5 ²	25 ²

relu	256	256
FC 2	256	100

УТОЧ: $364 \cdot 45^2 + 8592 \cdot \text{denom} + \text{leca} =$
 $= 364 \cdot 45^2 + 8592 \cdot 8 + 416 \cdot 256$

Bytes (S,B) = $364 \cdot 45^2 + 8592 \cdot 8 + 416 \cdot 256$

Memory ~~leca~~

Memory (S,B) = $\text{leca} + 52 \cdot 45^2$

Latency (S,B,0) = $t_0 + \text{mem} \cdot \text{as} \cdot \text{as} \cdot \text{as}$

Energy (S,B,0) = Latency (S,B,0) +

+ e-f FLOPS + e-m Bytes

где e-f - энергия преобразования на один шаг

e-m - энергия на чтение/запись